Multi-Source Candidate Data Transformer

Eightfold Engineering Intern Assignment

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1. Pipeline

Input Sources → Parsing → Validation → Normalization → Entity Resolution → Conflict Resolution → Confidence Calculation → Canonical Profile → Projection Layer → Output Validation. Deterministic and resilient to malformed inputs.

2. Canonical Schema

Candidate ID, full name, emails, phones (E.164), location (ISO■3166), links, headline, years of experience, canonical skills, experience, education, provenance and overall confidence. Dates use YYYY■MM.

3. Merge Strategy

Match candidates using email then phone. Resolve conflicts using source reliability (Recruiter CSV > ATS JSON > Resume > Notes), normalized equality and completeness. Skills, emails and links are union-merged. Experience records merge only when company, title, start date and end date all match.

4. Confidence

Confidence combines source reliability, validation success, normalization success, cross-source agreement and completeness. Every merged field records provenance, winning source and merge reason.

5. Projection

Runtime JSON configuration supports field selection, renaming, normalization, provenance/confidence toggles and missing-value policies while preserving the internal canonical model.

6. Edge Cases

Handled: malformed inputs, duplicate candidates, conflicting values, invalid phones/dates and partial records. Out of scope: OCR for scanned PDFs, distributed entity resolution and advanced NLP.